

Course Code: CCE-332,
Course Title: Peripheral and Interfacing Sessional

Group: 09

Comparison with related works:

Name	Primary Function(s)	Locomotion & Control	Key Features & Innovation
TrashBot (Our Project)	Autonomous waste collection in public areas and smart segregation of metallic vs. non-metallic waste	4-wheel differential drive platform controlled by a Raspberry Pi 3 Model B, utilizing HC-SR04 ultrasonic sensors for obstacle avoidance and an ESP-32 for IoT connectivity.	Features dual-mode operation. Uses a multi-degree-of-freedom robotic arm with MG995 servos to grab waste, and an inductive proximity sensor for automated segregation. Developed as a highly capable, low-cost prototype
The V Robot [1]	Indoor mixed waste collection and real-time autonomous sorting.	Wheeled, Raspberry Pi 4, webcam.	Uses OpenCV for real-time visual waste identification and a 3-joint servo arm for physical grasping. Demonstrates that practical segregation can be achieved on affordable hardware.
DustCart [2]	Door-to-door separate waste collection on demand in urban pedestrian zones.	Wheeled balancing robot, GPS navigation combined with pre-loaded maps, LiDAR/ultrasonic.	Operates on an on-demand basis where users summon it via phone call or SMS. Features a touchscreen interface for users to deposit waste, and simultaneously monitors urban air quality using embedded sensors.
Trombia Free [3]	High-performance, large-area outdoor street and industrial sweeping.	Fully autonomous, all-electric, 3D LiDAR, GNSS satellite positioning.	Uses a filter less cyclone dust control system and an automated emptying station for continuous, heavy-duty fine dust removal.
SmartBin Rover [4]	Mobile dustbin that patrols predefined routes and approaches users for waste disposal.	Wheeled, GPS, ultrasonic sensors, AI person-tracking (YOLO).	It responds to approaching people automatically to encourage proper waste disposal and reduce public littering.

References:

[1] "Waste collection in busy indoor spaces using everyday hardware," *International Journal of Creative Research Thoughts (IJCRT)*, Vol. 14, Issue 6, 2026.

[2] Ferri, G., et al., "DustCart, a Mobile Robot for Urban Environments: Experiments of Pollution Monitoring and Mapping during Autonomous Navigation in Urban Scenarios," *Proceedings of the International Conference on Robotics and Automation (ICRA)*, 2010.

[3] Trombia Technologies, "Trombia Free - Autonomous Street Sweeper," *Trombia.com*, 2026.

[4] "Review paper on AI-powered autonomous moving dustbin using person tracking and object motion prediction," *RJ Wave*, 2025.